

**Environmental
Monitoring System
Supply Specification**

Project No.:	01000
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1. GENERAL

The PACKING, MARKING AND SHIPPING INSTRUCTIONS specify the preparation, the packing and marking of all equipment, materials and miscellaneous items in accordance with the PURCHASE ORDER.

2. DEFINITIONS

GOODS	A generic term describing materials, equipment, products, supplies spares, documentation data, computer programs, labor(construction, design, maintenance, supervision), or other deliverables provided by a supplier. Whenever appropriate, items
SUPPLIER	The company or organization which supplies the goods and services
COMPANY	Karabatan Utility Solutions LLP
CONTRACTOR	a firm awarded with a contract by CLIENT to provide Engineering/Procurement for the POWER PLANT PROJECT
PROJECT	Karabatan Power Plant
COSHH	Control of Substances Hazardous to Health

3. SCOPE

The SUPPLIER shall prepare pack and mark all GOODS in accordance to this PACKING, MARKING AND SHIPPING INSTRUCTIONS.

4. GENERAL

Packing means to efficiently and effectively protect GOODS, in SUPPLIER's time schedule, for a period up to 12 months from EXW delivery. Packing shall be sufficient to withstand, without limitation, rough handling during loading/unloading, sea and/or inland transport, bad handling during transit, and outdoor storage. Depending on their type, GOODS are to be protected against both mechanical damage (shocks, rupture, breakage, loss or pilfering) and corrosion (rain, salty atmosphere, etc).

Equipment and materials must be export packed as per regulation of both export nation and Kazakhstan.

SUPPLIER is to verify and adhere at supplier's own cost, to the latest regulations applicable for Kazakhstan.

Packing list, packing note and marking of the packages shall be in English language at least.

5. RESPONSIBILITIES AND WARRANTIES

- SUPPLIER shall be responsible for compliance with these instructions and the choice of category for packing, according to the nature of the GOODS and transport facilities used.
- CONTRACTOR reserves the right to inspect the packing at any time, and to reject it when it does not meet packing requirements.
- SUPPLIER shall be responsible for the accuracy of the weights and dimensions declared, and marking of packages. The packing list shall be in strict conformity with the packing content.
- All costs incurred in connection with missing or damaged GOODS caused by non compliance with these instructions shall be charged to the SUPPLIER.
- After notification by CONTRACTOR, SUPPLIER shall take immediate action for the replacements of such missing or damaged GOODS within an acceptable time schedule, at SUPPLIER's own cost.
- SUPPLIER shall be responsible for recommendations for storage of the GOODS for 12 months according to site conditions. Any damage incurred due to faulty or missing recommendations shall be borne by the SUPPLIER.
- SUPPLIER shall be responsible for any loss or damage caused through wrong marking.
- Penalties would apply due to the late delivery, resulting from defective marking.

6. PREPARATION FOR PACKING

Prior to packing, GOODS shall be protected under SUPPLIER's responsibility as hereunder described and not limited to:

6.1. GENERAL

- 6.1.1.** The GOODS shall be dried, cleaned internally and externally, removed from all welded spatter, scale, rust, cuttings, fittings, etc as well as any and all foreign material.
- 6.1.2.** Protection against environmental corrosion caused by extremes of humidity, condensation, rain, dust, sand, salted air and sea spray shall be provided. All treated items are to be fitted with a red tag stating Caution – This equipment has been treated with a rust preventative and is not to be disturbed until ready for service. If protective

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coating is damaged it must be re-instated for continued storage.

- 6.1.3.** Critical wearing surfaces, highly stressed parts, and highly finished surfaces shall be further cleaned by dipping or brushing with a suitable product, such as petroleum naphtha or alkaline cleaning compound. Close tolerance surfaces shall be additionally cleaned to ensure removal of perspiration and fingerprint stains, using any commercially available oil fingerprint remover.
- 6.1.4.** The GOODS being delivered by partial unit or partially dismantled shall be furnished with suitable markings required for assembly and installation at the site.
- 6.1.5.** Unless otherwise specified, after hydro-testing, operation, and/or performance test, all fluids, i.e., lubricating oil, fuel oil water, etc., shall be completely drained from all tanks, vessels, jacketing, piping, etc., and wiped or blown dry. Lint free rags are to be used for wiping.
- 6.1.6.** When hydro-testing stainless steel equipment and materials, precautions shall be taken to prevent pitting corrosion and stress corrosion cracking.
- 6.1.7.** When the packing is carried out in boxes and/or crates, the metal or plastic corners at the slinging points shall be used.
- 6.1.8.** The piping material will be packed as per ATTACHMENT 3 PACKING OF PIPING MATERIAL.
- 6.1.9.** Specific points due to the requirements of the destination, terms of the purchase order etc will be included in the present instructions in due time.

6.2. SMALL LOOSE PARTS, TOOLS AND SPARES PARTS

Spare parts, accessories or whatever should be transported together with main equipment within one complete package (no partial shipment) in order to enable the COMPANY to import into Kazakhstan on duty free basis.

6.2.1. Small Loose Parts

Small loose parts shall be tagged and packed in a separate, sturdy wooden box. Any such box shall be shipped with main equipment, being securely attached to the main skid or within the main box. Secure attachment consists of steel strapping or bolting. Tiring or wiring is prohibited.

6.2.2. Special Tools

Special Tools shall be packed in an individual, rugged wooden box and marked as Special Tools.

6.2.3. Spare Parts For Erection, Commissioning and For Normal Operation

Each category of spare parts shall respectively be packed in an individual rugged wooden box and marked as Spare Parts.

6.3. CERTIFICATE OF PACKING

6.3.1. All goods imported from EU, U.S.A., Canada, Korea, Japan and China, supplier should be required to submit one of the following type of original, signed packing certificates,

- Fumigation Certificate
- Non-Wood Packing Declaration
- Declaration of Non-coniferous Wood Packing Material (especially for the package coming from USA, Japan, Korea & Canada)

and should be in compliance with following requirements

- Supplier shall submit the official Fumigation Certificate, in which disinfect means, doses of treatment, time and temperature of treatment declared, issued by national plant quarantine authority of exporting country when cargoes imported from above mentioned countries contains wooden packing materials. If cargoes without wood packing material a Non-Wood Packing Declaration offered by supplier, which is to certify this cargo does not contain wooden packing materials, should be submitted.
- The treatments are accepted by Kazakhstan include fumigation with Methyl Bromide and heat treatment. The Kazakhstan side will approve other quarantine treatments that assessed and proved by Kazakhstan experts as effective quarantine treatments, if the technical data of experiment was submitted to Kazakhstan side by the plant quarantine authority of exporting country in advance.
- Before cargoes depart from port, the disinfections treatment to wood packing material should be conducted under the supervision of the official quarantine authority of the exporting country if cargo conveyed by container is fumigated with Methyl Bromide, the residual in container must be below the safety level, i.e. 5ppm before departure from the exporting port.
- Each of wood packing materials should be marked after being disinfected. The type of marking of wood packing is refer to instruction in ATTACHMENT 4. But the method, place, time and the operating body of quarantine treatment or the code of the said body should be indicated in this distinguished mark.
- Goods, carried by containers, containing wood packing materials from the 3rd nation or areas of origin transshipping through above countries to Kazakhstan, may not be covered by above-mentioned requirement when goods, carried by containers, containing wood packing materials was not unloaded and replaced its wood packing materials in above countries. However the goods are affected

if the goods are processed, assembled or repacked by wood package in above countries in accordance with above requirement.

- The goods containing wood packing materials from above countries is covered by above requirement if its wood package not replaced in the 3rd nation or area. If wood package of the goods replaced in the 3rd nation or area, a certificate, which attests the wood packing used in is corresponding with the quarantine regulations of Kazakhstan, should be issued by the official quarantine authority of the 3rd nation or area.
- Wood packing materials are defined as used or for use with cargo to prevent damage, including, but not limited to, dunnage, crating, packing blocks, drums, cases, spools and skids.
- The member nations of the EU includes: Austria, Belgium, Croatia, Check Republic, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, Netherlands, Portugal, Slovakia, Spain, Sweden, and Britain.

- 6.3.2.** Wood materials used to bear, pack, support, strength, bedding, etc. are liable for such compliance.
- 6.3.3.** The original Fumigation Certificate, Non-wood Package Declaration or Non coniferous Wood Packing Material must be made available to COMPANY or CONTRACTOR prior to the goods arrival in Kazakhstan.
- 6.3.4.** If the packing certifications do not meet the Kazakhstan authorities' requirement, SUPPLIER shall take all necessary actions including fumigation of all the goods to ensure the requirements are met.
- 6.3.5.** All costs to comply with the above shall be the responsibility of SUPPLIER.
- 6.3.6.** In case any changes of regulation in Kazakhstan, SUPPLIER shall comply with the requirements immediately.

7. HEAVY LIFT AND OUT OF GAUGE EQUIPMENT

7.1. DIMENSIONS AND WEIGHT

Equipment shall be classified as OUT OF GAUGE or HEAVY LIFT, when its characteristics exceed one of the following criteria:

Length : 25 Mtr

Width : 3.5 Mtr

Height : 4.3 Mtr

Weight : 22 Tonnes

7.2. INTERNAL SUPPORT

For large size of equipment, such as, columns, reactors, pressure vessels, stack/chimney etc., SUPPLIER shall provide supports to prevent deformation.

7.3. LOAD DISTRIBUTION

As far as transport drawings issued by SUPPLIER are confirmed, SUPPLIER shall control the adequacy of the supports and/or saddles and ensure that load distribution now fixed at 4 tonnes/m² (to be confirmed in due time), is in compliance with CONTRACTOR's nominated Freight Forwarder requirements with respect to road, rail and sea transportation and unloading port requirements.

In case SUPPLIER has the responsibility to deliver on a DDU site basis, SUPPLIER shall take into consideration, prior to saddle fabrication the maximum load distribution allowed with respect to local means of transportation and handling and port requirements.

In case of CIF nominated port delivery term basis, SUPPLIER shall liaise with CONTRACTOR's /COMPANY's nominated Freight Forwarder with respect to local load distributions tolerance prior to saddle fabrication.

7.4. PREPARATION OF SEA FASTENING OPERATIONS

For any contractual delivery terms stated on the purchase order (EXW, FOB, CIF etc.), the SUPPLIER shall carry out the preparation of sea fastening according to the instructions which will be given by the shipping company through CONTRACTOR's freight forwarder and the following:

The unpacked equipment shall be prepared for sea fastening operations and be equipped as follows:

- A. Lifting lugs to be calculated, defined and shown on the transport drawings. They will be used for any handling operations onto trucks for road transportation and onto vessels for sea transportation and onto wagon for rail transportation; wireguard included.
- B. Lashing points shall be calculated in order to support all forces crosswise, lengthwise, and so on They will be clearly marked on the equipment and transport drawings.
- C. When the specificity or equipment required utilization of special tools, such as lifting beams, slings, etc for handling operations, the SUPPLIER shall prepare a list of special tools which will be proposed for CONTRACTOR's approval and imperatively included in the Purchase order.

The SUPPLIER shall be responsible for the non-observance of the above.

8. CONTAINERIZATION

At SUPPLIER's cost, the SUPPLIER or their packer shall carry out the stuffing operations of goods into container when required.

9. MISCELLANEOUS

Supplier should consider the following points as priorities.

9.1. Preservation

9.1.1. Rust prevention

- All unpainted machined surfaces, both external and internal shall be cleaned and dried as specified above, and amply coated with a rust preventative, which will maintain a protective coating for a minimum of twenty four months, so long as the coating is not burned, dissolved or mechanically rubbed off.
- For machined surfaces which are external, the rust preventative shall be a heavy application of grease or liquid film which dries to a tough coating and which conforms with the performance requirements listed below:

Application method – Hot application, brush or dip, maximum temperature of application 95° C.

Description and use – Thick, hard drying, greasy film. For protection of highly finished parts of simple design, for long periods.

- Acceptable preservatives are available from the major oil companies and their dealers, for example:

Gulf Gulf No-Rust No 1

Esso (Enjay Chem) Rust Ban 385

Shell Ensis 264

Mobil Mobilcoat601

Texaco Texaco Rust proof Compound L

Valvoline Tectyl 506.

Other brands are available and are acceptable providing they conform to the performance requirements above.

- For machined surfaces which are internal to an assembled unit, the rust preventative should be of a type which does not require removal before operation of the equipment and conforms to the performance requirements below:

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Application method – Brush, dip or spray, 21°C to 24°C.

Description - Engine lube oil with corrosion inhibitors.

- Acceptable preservatives are available from the major oil companies and their dealers, for example:

Gulf Gulf No-Rust Engine Oil

Esso (Enjay Chem) Rustban No 623

Shell Ensio Engine Oil Grade 20

Mobil Mobilcoat 501 or 503

Texaco Preservative Oil 10W

Other brands are available and are acceptable providing they conform to the performance requirements above.

- Shipping by sea on deck is not appropriate for stainless steel items, e.g. large vessels or heater tubing modules. If this is unavoidable the items must be sprayed with a suitable protective coating e.g. Petrolite's Kontol KP-94, which is to be removed at site with hydrocarbon solvent or detergent solution.
- Acceptable application methods are spraying, brushing, filling etc through all openings or by disassembly if necessary. All internal unpainted surfaces are to be completely coated. Particular attention to be given to journal surfaces of sleeve bearings and complete anti-friction bearings, to ensure they are completely coated.
- All treated items are to be fitted with a red tag stating "Caution This equipment has been treated with a rust preventative and is not to be disturbed until ready for service. If protective coating is damaged it must be re-instated for continued storage ."
- Piping disassembled from equipment shall have internal surfaces free from scale and foreign matter and coated with a rust preventative as per the above requirements. Pickling and cleaning is also acceptable. Pipe ends shall be protected use of plastic caps or wrapped to prevent ingress of dirt and moisture.
- Exposed machine parts shall also be wrapped with heavy waterproof paper or plastic c/w vapor corrosion inhibitor crystals. Sharp edges to be removed to prevent puncture of protective cover.

Spare parts are to be similarly protected and have an additional second layer of wrapping.

Care is to be taken to prevent damage by the protective coating to items made from rubber, mica and similar material.

9.1.2. Protection of openings

- Flanged openings shall be protected with bolt-on wooden covers (6mm min) or metal covers and gaskets using at least 3 bolts.

Wiring of covers is not acceptable. Machined flange surfaces to be coated with a suitable rust inhibitor before blanking.

After sealing with blanks, the joint is to be sealed with waterproof tape such as Densotape.

- Exposed pipe threads shall be protected with threaded metal plugs or caps, identified with metal tags explaining purpose. Supplier is to provide drawings identifying connections/ no s.

Plastic covers/ thread protectors are prohibited, unless prior written approval is given by CONTRACTOR/COMPANY.

All loose covers, etc shall be tightly secured.

9.1.3. Flexible connections

- All flexible connections subject to transit damage are to be removed and packed in a separate plywood box securely fixed to the main case/ crate.
- Bellows retaining bars and shields are to be left in place for removal at site. Protective bars/
- shields are to be painted yellow to signify temporary items.

9.1.4. Shaft couplings

Flexible couplings are to be broken and adequately supported/ protected to prevent damage. Rigid couplings shall be protected in accordance with manufacturer s recommendations. Membrane spacers etc shall be securely packed in a separate plywood case fixed inside the parent case/ crate.

9.1.5. Dehydration

- Use of desiccant is particularly important for enclosed equipment subject to wide temperature/ humidity changes over a 24 hour period e.g. during ocean transport. Desiccant crystals are to be uniformly distributed over the equipment to provide adequate protection.

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- Quantity of desiccant shall be determined on the basis of surface area of the moisture barrier, moisture vapour transmission rate and volume of the enclosed space. Additional desiccant may be required where moisture absorbent materials are used in the packing. Desiccant shall be of the type that indicates the presence of moisture by colour change. Silica gel in crystal or pellet form only, is to be used. Any alternative is to be approved by CONTRACTOR/COMPANY.
- Desiccant filled porous bags are to be hung on the inside of covers of major connections for rotating equipment, unless physical dimension is restrictive.
- All equipment shipped using dehydration materials shall be labeled or tagged with a high visibility red sign reading – “Caution – desiccant materials enclosed. Do not operate equipment before removal. If integrity of the closure is compromised desiccant must be replaced for continued storage ”
- Supplier shall provide COSHH data for materials used.

9.1.6. Painting

Items to be painted as per specific contract requisition/specifications.

9.2. STATEMENT COVERING PACKING OF EQUIPMENT

9.2.1. For Standard Packing

- The type of packing shall be determined by the Supplier and may consist of boxing or crating, bundles, bags, moulded polystyrene, fiberboard boxes, barrels, metal drums or merely skidded. The selection should be based on suitability of the type of material to be packed and method of shipment.
- Where pallets are used, standard size pallets/ and or pallet sized packing cases shall be preferred to facilitate handling by fork lift trucks at all stages from Supplier to site.
- Carbon steel pipe-work up to 75mm N.B. shall be oiled and packed in open crates with solid timber ends.
- Carbon steel pipe-work over 75mm N.B. and up to 150N.B shall be oiled and bundled with steel banding.
- Carbon steel pipe over 150mm N.B. may be oiled and dispatched loose. All pipe ends are to be capped or wrapped to prevent damage/ ingress of dirt and moisture.
- Large volume equipment such as pressure vessels, large machinery pieces, structural steel, large size pipe etc., which are inherently weather resistant and

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capable of withstanding the rigors of transport, may be shipped without being boxed.

- Electric cable shall be supplied on drums suitable for overseas shipment, with ends sealed to prevent ingress of moisture. Cable drums shall be suitable for storage in the direct sunlight.
- All vessels and skid units shall be fitted with strong wooden or metal skids or cradles corresponding in size to the weight of the equipment. These skids or cradles shall be bolted or strapped to the vessel or skid unit to enable it to be free standing. A double layer of wax paper shall be applied between the steel straps and vessel to prevent the straps from touching the vessel.
- Cases and crates exceeding 500kg shall be fitted with lifting plates and attachments to facilitate lifting by steel cables. Lifting lugs shall be clearly identified by stencil marks.
- Use of steel shipping containers should be considered for lightweight or standard packing, particularly when arrangements can be made for a bonded warehouse/ compound at site to eliminate the need for customs clearance at the port of entry, possibly avoiding the need for intermediate unpacking.
- Any equipment containing stainless steel shall be cased and marked "This material to be stored below deck and not as deck cargo".
- Cases, crates and other packing materials shall be considered to be expendable items and non-returnable.
- Where Supplier considers packing materials to be returnable (e.g. cable drums), the Supplier shall state so clearly in his quotation, the requirement for return of the materials and the additional cost implication of non return. CONTRACTOR/OWNER shall then decide the most cost effective option.

9.2.2. For More Robust Packaging

- Very large and complicated machinery such as compressors, generators, large motors, engines etc, not supplied with self supporting base-plates, shall be bolted to a skid formed of heavy timbers suit able for all types of handling. This type of equipment shall be wrapped with heavy tarpaulin, or plastic so as to be enclosed on all sides except underneath, to enable free circulation of air. Covering shall be secured to the skid or base-plate in such a manner that it will not extend beyond the edge or be torn by forks etc.

- Pressure vessels shall be supplied with saddles to facilitate transport and ease of lifting. Horizontal vessels shall have permanently fixed (welded) plates saddles supplied as part of the structure.
- Vertical vessels/ columns shall be supplied with suitable temporary transport saddles, which are to be fixed by means of steel bands, or welded to transport cleats supplied on the vessel. Saddles shall have adequate ground clearance to facilitate lifting at site.
- All other equipment shall be enclosed in nailed wooden cases or crates with steel strapping. Cardboard cartons shall not be used for packaging.

9.2.3. For Wooden Cases and Crates

Wooden cases and crates shall be sufficiently strong to enable stacking and comply with the following requirements:

- All cases and crates shall be constructed of new timbers of not less than 22mm thickness (or 10mm for plywood), and shall be lined with high quality waterproof bitumen lined paper.
- Case lids shall be of tongue and groove timber or plywood, lined as above.
- All cases shall be fitted with strong wooden skids corresponding in size to the weight of the case but not less than 75mm square.
- Where rubbing strips are required they shall be supplied not less than 25mm thick along the top outer edge of the cases. Metal strips of a suitable thickness can also be used if more appropriate.
- Where wood comes into contact with any part of the equipment, it shall be covered with waxed paper to prevent ingress of moisture penetrating to metal parts.
- Boxes are to be fitted with screened vents suitable for the size of box. Screens are to be located on side walls and have internal splash boards so as to prevent ingress of water and allow circulation of air to prevent sweating from temperature fluctuations.
- All wood shall comply with requirements for treatment of woods imported to Kazakhstan.

All instrument panels, instruments, electrical equipment, precision machinery and control valves shall be packed in cases constructed from tongue and groove timber or plywood. They shall also be vacuum heat sealed in a polythene barrier c/w sufficient silica gel to afford protection for 18 months.

Non precision equipment can be packed in cases constructed from butt jointed timbers or plywood, lined with bitumen paper.

Where possible all major equipment shall be fastened to the base of the packing case. Where this is not possible the equipment is to be blocked securely with timber to prevent movement during transport.

10. MARKING INSTRUCTIONS AND STORAGE CODE

Marking for each package shall be in accordance with pattern in ATTACHMENT 1 MAIN SHIPPING MARK.

10.1. TYPE AND SIZE OF THE GRAPHICS

Packages and cases will be marked with indelible black paint resistant to sea water. Marking must be perfectly legible and with the contrasting color.

Ex: If the background to be marked is “black” or dark color, the marking should be “white”.

The marking will be in English language.

Stencil on two opposite sides and the top of each package in clear characters at least 40 mm high (where case size permits, otherwise use optimum size for each package dimension).

N.B.: - Metal tags or labels must be stamped or indelibly marked with full shipping marks and must be securely attached, welded or riveted to all loose, bundles or uncrated items.

- Cable drums: On the two flanges.

(as per indication in ATTACHMENT 4)

Dimensions of cases or packages will be indicated in cm and the weight in kg . No other units shall be used.

10.2. COLOR CODE

Each package, bundle item or container shall be identified by a common blue color.

- This color identification shall be accomplished by application of weather resistant, paint to corners of each package or container.
- Color band will be used for unpacked material such as pipes, steel structure, drums etc.

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10.3. STORAGE CODE

The type of storage code to be printed on the two opposite sides and top of each package is specified as follows:

- X - Crates or Packages to be stored outside
- XX - Crates or Packages to be stored under cover / or open shed
- XXX - Cases or Packages to be stored in covered or closed premises
- XXXX - Cases or Packages to be stored in air conditioned closed premises

Refer to Equipment Storage Codes in ATTACHMENT 2.

Supplier advises whether the equipment being supplied requires special storage facilities (e.g. air conditioned). Supplier provides instructions for on site maintenance by COMPANY and ensure the serviceability of the equipment and maintain the validity of the equipment guarantee, including special reference to equipment expected to be stored in excess of 12 months.

10.4. PACKAGE NUMBERING

The package number within the main shipping mark of each package shall be as follows:

PACKING LIST NUMBER

CONTRACTOR No.	Traffic Control Number (T.C.N.) assigned by CONTRACTOR Expediting group for a complete shipment, corresponding to the Release for Packing No.
----------------	--

PACKAGE No.	1 / n
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The first number being that of the order in the batch, then after oblique, the last number being the total in the batch (i.e. the total of packing list). For example, for a batch of 7 packages:

First package CONTRACTOR No.

Package 1 / 7

Last package CONTRACTOR No. (same number)

Package 7 / 7

10.5. INTERNATIONAL STANDARD SYMBOLS

10.5.1. For Handling

All boxes and crates shall also be marked with the appropriate international standard symbols for handling.

All boxes and crates are to be marked clearly on all low sides:

“HANDLE WITH CARE”

and

“RIGHT SIDE UP”

In addition, each box shall be marked “KEEP DRY” on all four sides.

In the case of packages with a single gross weight totaling 2,000 kgs and/or a height of more than 1 m, the centre of gravity shall be clearly marked with the symbol on two adjoining sides.

For all items of equipment with an eccentric centre of gravity this symbol shall be marked at the bottom, side and top of the package.

The slinging points shall be marked with a chain symbol.

10.5.2. For Dangerous Goods

Packages containing goods as per statutory regulations on the packing and transport of dangerous goods are to be marked on all side with appropriate labels in conformity with IMDG/ADR/RID/IATA regulations.

N.B.: incorrect marking:

It is expressly understood that the SUPPLIER shall be responsible for all expenses caused by improper or wrong marking contents.

11. PACKING LIST AND PACKING NOTE

11.1. PACKING LIST

11.1.1. The packing list will be prepared on CONTRACTOR's standard forms and will show the summary of the content of each package.

11.1.2. Numbering of Packing List

- Only one packing list number shall provide by CONTRACTOR's Expediting group for each definite batch of goods.
- This number is named T.C.N. (Traffic Control Number).
- This T.C.N. will be marked on the package and on the shipping documentation and used as unique number to facilitate the tracking.

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- The same number will be shown on the Shipping Release Notice issued by CONTRACTOR's Expediting group expediting when the batch is technically accepted.

11.1.3. Two (2) copies of the packing list in waterproof plastic envelope must be securely nailed under a steel sheet, on the outer surface of the first package of the batch with the associated packing note corresponding to the package, as per indication in ATTACHMENT 4.

11.2. PACKING NOTE

11.2.1. The description of the packing note must correspond exactly to the detailed contents of the package. The expression "accessories for" or "batch of nuts and bolts" for example, are formally forbidden.

11.2.2. One copy of the packing note shall be placed in a waterproof plastic envelope inside of each package. Another copy, in a waterproof plastic envelope shall be placed outside of the package under a steel sheet as per indication in ATTACHMENT 4.

11.3. LANGUAGE OF PACKING

Above mentioned shipping documentation shall be in English Language at least.

12. PACKING INSPECTION – PACKING PERFORMANCE

The packing inspection will be confirmed by CONTRACTOR in due time. It shall take place at SUPPLIER's designated premises, as per the following procedure:

- Cases and/or crates shall be left open.
- Shrink wrapped plastic covering or vacuum plastic covering operations to be in standby.
- Packing lists and packing notes to be handed over to entitled inspector,

For marking control

For compliance examination of cases contents

Consequently, SUPPLIER shall:

- After technical acceptance of equipment by CONTRACTOR, define the expected necessary packing preparation period.
- Prior to scheduled packing inspection date, provide CONTRACTOR's Expediting group with packing lists and packing notes.
- Advise by fax CONTRACTOR Expediting group, stating expected date and location of packing acceptance to be performed by CONTRACTOR's representative.

- After packing acceptance notified by CONTRACTOR, proceed to :
 - Shrink wrapping operation as well as vacuum sealing,
 - Closing of cases and/or crates together with incorporation of the appropriate packing lists and packing notes as per Article 10.0 here above.

13. SHIPPING INSTRUCTIONS

13.1. DOCUMENTATION

One set of installation instructions and drawings shall be included with the shipment, in a sealed waterproof envelope, firmly attached to the equipment inside the case or crate.

SUPPLIER shall provide CONTRACTOR Expediting group with the following:

- 13.1.1.** Two (2) weeks after the issue of purchase order, three (3) copies of preliminary packing list as per CONTRACTOR/COMPANY standard forms including the number of packages with their respective dimensions and weights evidencing storage code.
- 13.1.2.** Five (5) months before the contractual delivery date of the equipment, three (3) copies of preliminary transportation drawings for heavy or over dimensional packages where weights and/or dimensions are equal or superior to:
 - Weight: 22 Tonnes
 - Length: 25 Mtr
 - Width: 3.5 Mtr
 - Height: 4.3 Mtr

(the above dimensions and weight will be confirmed in due time)

with the following indications :

- Three planes view of equipment,
- Actual position of all nozzles and protruding parts,
- Net and gross weights,
- Position of the center of gravity on the three planes,
- Accurate size and overall dimensions of cases, crates or packages,
- Slinging points of equipment or package in all planes.

Support positions and dimensions shall be indicated in case of unpacked material and a calculation sheet for support may be required for very heavy packages.

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The transport drawings will be verified by the freight forwarder named by the CONTRACTOR. They will be considered final when they will have obtained approval by the freight forwarder.

13.1.3. Seventy (70) days before the expected date of availability of goods for shipment, SUPPLIER shall:

- Confirm by fax to CONTRACTOR Expediting group that the preliminary packing list and / or transportation drawings are unchanged or,
- If modified, airmail or courier to CONTRACTOR Expediting group such revised documents in three (3) copies.

13.1.4. Prior to effective date of inspection if any, by CONTRACTOR's representative, SUPPLIER's documentation but not limited to the following shall be available in CONTRACTOR's office:

A. Packing list

Five (5) copies. To be prepared on CONTRACTOR's/COMPANY's standard form.

B. Packing note

Five (5) copies. To be prepared on CONTRACTOR's/COMPANY's standard form.

C. Proforma invoice

Five (5) copies of SUPPLIER's proforma invoice shall be addressed to CONTRACTOR/COMPANY, evidencing following information:

- Number of delivered items (complete or partial or balance according to Purchase Order),
- Quantities of sold goods with currency used in the Purchase Order,
- Goods commercial designation as per requisition,
- Weights/volume,
- Unit and total value,
- Place and conditions of delivery.

A particular attention shall be brought to HAZARDOUS GOODS which shall be identified on a separate invoice and packing list to those for non hazardous material.

All hazardous material shall be identified by the appropriate hazard class and technical or proper shipping name.

D. Certificate of dangerous goods (when applicable)

Three (3) copies of such certificate shall be transmitted to CONTRACTOR.

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E. Certificate of origin (when required)

SUPPLIER shall provide CONTRACTOR with a certificate of origin stating the country of origin for materials manufactured outside of E.E.C., duly legalized by local Chamber of Commerce.

For the materials manufactured inside E.E.C., SUPPLIER shall provide CONTRACTOR with an attestation of origin with indication of manufacturing place.

F. Any documents, required by the final destination and by the specificity of the purchase order, defined in due time, shall be provided by the SUPPLIER.

13.2. FREIGHT FORWARDER

13.2.1. CONTRACTOR's/COMPANY's designated Freight Forwarder is to be nominated in due time;

The Freight Forwarder is the sole entity intended to perform the transport operations according to the contractual delivery terms.

In case of EXW or FCA delivery terms:

- Call forward of the GOODS ready to be shipped,
- Perform export clearing operations (EXW Only)
- Freight booking on appropriate mean of conveyance according to the packing list provided as per above paragraph 11.1
- In case of FOB delivery terms :
- Call forward of the GOODS indicating place and period of loading, name of the vessel and/or other means of transportation,
- Freight booking on appropriate mean of conveyance according to the packing list provided as per above para. 11.1,
- N.B. = Only when the CONTRACTOR has received and agreed the above defined documentation, will be issued the release for shipment notice notifying the date of collection of GOODS.
- The shipping instruction will be completed, in due time, in accordance with the requirements of the contract and the contractual terms of the Purchase Order of the SUPPLIER.

14. ATTACHMENTS

ATTACHMENT 1 MAIN SHIPPING MARK

ATTACHMENT 2 EQUIPMENT STORAGE CODES

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ATTACHMENT 3 PACKING OF PIPING MATERIAL

ATTACHMENT 4 POSITION OF SHIPPING-CARE MARK

ATTACHMENT 5 EXAMPLE OF TRANSPORT DRAWING

ATTACHMENT 1 MAIN SHIPPING MARK

MAIN SHIPPING MARK

CONTRACTOR :

CONTRACT TITLE :

POINT OF ENTRY :

CONSIGNEE :

SUPPLIER NAME :

P.O. NO. :

ITEM NO. :

T.C.N. :

COUNTRY OF ORIGIN :

PACKAGE : ** */* **

NET WEIGHT : KGS

GROSS WEIGHT : KGS

MEASUREMENT

LENGTH :

WIDTH :

HEIGHT :

In case of airfreight shipment the above mentioned, specifications remain the same, but instead of point of entry the marking shall be:

AIRPORT OF DESTINATION:

FOR DELIVERY TO: (refer to Consignee)

Two copies of the packing lists are to be attached to the outside of each package in waterproof plastic cover underneath a nailed plate. In the case of equipment which is

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dispatched unpacked, this plate should be welded on the metal parts (spot welding) or attached permanently to the package in another way depending on the material. One copy of the packing list is to be placed in a clearly visible point in the boxes/crates in a waterproof plastic cover.

ATTACHMENT 2 EQUIPMENT STORAGE CODE
EQUIPMENT STORAGE CODES

DESCRIPTION	STORAGE IN AIR CONDITIONED WAREHOUSE XXXX	STORAGE IN WAREHOUSE XXX	STORAGE OUTDOOR WITH COVER XX	STORAGE OUTDOOR WITHOUT COVER X
<u>Vessels</u>		Unpacked Spare parts, bolting, gaskets, etc. Packed Refractory lining cements	Tower and vessels internals Cyclones and component parts for field fabricated equipment	Unpacked Towers, Reactors, Regenerators, Converters, Drums, Storage Tanks, Vessels Packed Trays and internal components
<u>Heat Transfer</u>	Electrical ignition equipment control and instrumentation	Unpacked Test equipment and spares; test rings gaskets, etc.	Unpacked Heat exchangers of all types, Fired heaters, Furnaces, Stills, Rotary kilns, Stacks and flares, ducts, incinerators, steam generators, boilers of all types, surface condensers, ejectors, vacuum equipment, package refrigeration systems, deaerators, desuperheaters cooler boxes, vapourisers, tank heating coils and component parts for field fabricated equipment. Packed or Palletised Air cooler bundles, fans motors, gear boxes, louvers, actuators and cooling tower components	
<u>Mechanical Equipment</u>		Packed until required for assembly and/or installation Auxiliary equipment and spares, Lubrication systems, including auxiliary equipment, electrical generators, including drivers and ancillary equipment, Electric motors, Electrical and ignition equipment. Replacement cartridges	Pumps of all types Compressors of all types exhausts and blowers, including ducting, filters, silencers, dampers and controls Fans Mixers Reducers	Packed until required for assembly and/or installation Pulsation dampers, gas and diesel engine drives for mechanical equipment, Filters and separators of all types, heating, ventilation and Air conditioning equipment, Steam gas and hydraulic turbines and gas expanders, agitators and rotating disk contactors, decoking systems hoses and bulk handling equipment, finishing and packaging equipment.
<u>Construction</u>		Packed General consumables, protective clothing,		Packed Fuels and lubricants (in separate compound)

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		radiographic and stress relief materials. Unpacked Scaffolding fittings Welding rods and wires etc.		Unpacked Scaffolding tubes and boards Cutting and welding gases (in separate compound kept fully upright full and empty)
<u>Civil</u>		Packed Lubricant or Teflon coated slide plates, spring hangers		Unpacked Materials for: earthworks, roads, fencing, concrete aggregates, rebar & mesh, formwork, drains & underground piping, including fittings and specials, structural steelwork, platforms, ladders & handrails, building frames and cladding, railroad spurs, weight scales, elevators, trolleys, cranes & hoists, Bricks, concrete blocks, cable, cover
<u>Instruments</u>	Control room panels, Racks, cabinets and Associated instruments	Packed All types of instruments: Temperature, flow, level, pressure, analyzer etc. All small control valves Other instrument valves Panel boards, console desks. Instrument piping and tubing, including valves, fittings, nuts bolts, gaskets etc. Junction boxes relay boxes, instrument support and racks. Ambient heated or air conditioned storage areas.	Packed (Drummed) All types of instrument cables, mounted on cable drums (single and multi core)	Packed (Drummed) Instrument Housings 3" and over control valves
<u>Electrical</u>	Unpacked Non-metallic cable ducting, concrete earthenware, fiber and plastic. Large power transformers Power equipment, including HV, MV, LV switchgear, relays and relay panel etc. Notes: Internal covered storage in heated or air conditioned ancillary warehouse of required, due to ambient conditions prior to installation in substations or switch houses.	Unpacked Metallic conduit, trunk, trays for underground and aboveground power & lighting, including fittings. Lighting fixtures, receptacles, switch panel boards, lighting sockets, lighting wire and flexible cables. Communication equipment for telephone radio etc. Packed (Drummed) Instrument electrical cabling etc. Packed (Palletised) Electric tracing materials Power suppliers including battery chargers and voltage regulators.	Packed (Drummed) All types of electric power, lighting, and communication cables, wound on cable drums Packed (Palletised) Pre-cast concrete cable markers, tiles or bricks	
<u>Miscellaneous</u>		Packed Maintenance equipment; Lathes, drills, mills etc. Office and laboratory equipment, Furniture, filling cabinets, etc. Safety equipment, Fire blankets Testing equipment Radiographic, catalysts and chemicals. Spare parts for pressure	Packed Package units including desalters, electro static precipitators, air separation units, low temperature purification units, inert gas generators, process dryers, instrument air systems, chemical injection systems. Protection equipment including, fire hydrants, monitors, fire fighting vehicles, eye baths and showers	

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		vessels, heat transfer equipment civil piping, instruments, electrical		
<u>Protective cover</u>		Packed (Crated) Non-asbestos containing insulating blankets, pre-formed blocks, non-metallic cladding and underground coating/wrapping materials. Packed (Canned) Paints (in separate fireproof store) Packed (bagged) Refractory cements (in separate refractory store) Unpacked Refractory aggregates (as above)		Packed (Crated) Metal cladding for pipe and equipment insulation Unpacked Aggregates, bricks, tiles and re-inforcing mesh for heat or corrosion resistant linings or paving
<u>Piping</u>		Valves (All 6" and less) CS & Alloy Pipe (All 1.5" and less) Other Metallic Pipe Non-Metallic Pipe (PVC & Plastic) Flanges & Fittings (All 10" and less) Bolts, Studs & Gaskets Piping Specialties Other Specialties	Remaining pipe, valves, fitting and flanges	Pipe Supports

ATTACHMENT 3 PACKING OF PIPING MATERIAL

A. PIPES (CARBON STEEL) From 1/2" to 4"

1.PACKING IN CRATE - CATEGORY 3

WITH SOLID PANEL SIDES AT THE EXTREMITY OF THE CRATE

1.1 Physico-chemical protection :

All parts vulnerable to corrosion must be protected by the manufacturer, who will in particular take responsibility for draining, rinsing, drying, protecting and for other specific and adapted cathodic protection.

1.2 Mechanical protection

The pipes will be immobilized by appropriate inserts fixed to the crates.

1.3 Marking

The contractual shipping marking will be fixed to the sides of the crate.

B. PIPES (CARBON STEEL) 4" to 8"

1. BUNDLE - CATEGORY 2 - (refer to sketch annex 1)

1.1 Physico - chemical protection

The pipes shall be provided with initial protection carried out by the manufacturer such as protective oil and internal protection. This protection comprises a spray - applied protective and sealed ends (waterproofing, wood, metal or plastic covers) or any chemical protection perfectly adapted to the pipes proposed by the manufacturer and approved by Contractor.

1.2 Mechanical protection

The bundling is carried out with U - IRON or crosswise planks, joined with threaded connecting rods with stop - nuts. Quantities and strapping positions depend on the lengths, with a 120 cm spacing to prevent distortion.

C. PIPES (CARBON STEEL) 8"

1. LOOSE

ENDS TO BE PROTECTED WITH PLASTIC COVERS

1.1 Physico - chemical protection

Refer to above para 2.1

1.2 Marking contract or label affixed to the bundles.

D. STAINLESS STEEL PIPES (AND OTHER STEEL NOT ALREADY INDICATED) FITTINGS AND FLANGES STAINLESS STEEL AND OTHERS

1. CASE - CATEGORY 4

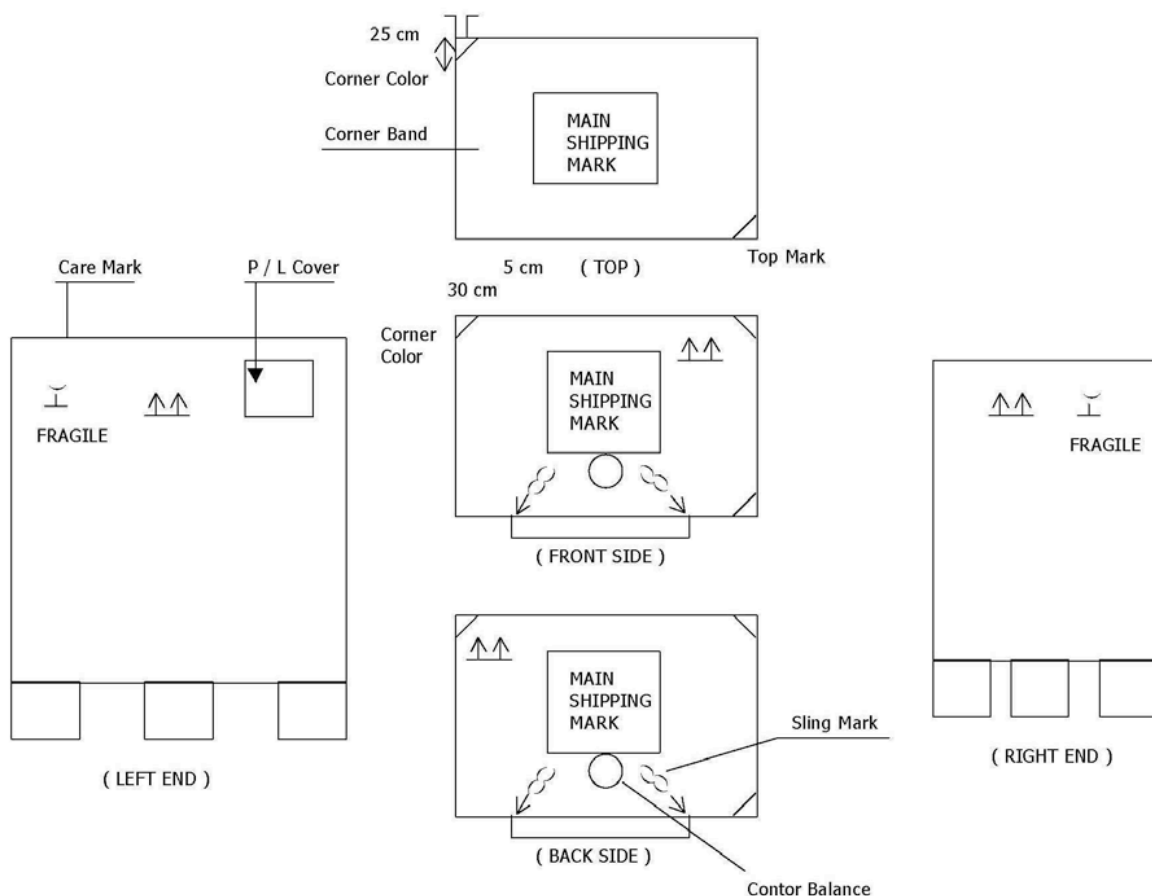
1.1 Physico – chemical protection

- If needed for this type of pipes, it will be carried out by the manufacturer
- All parts of fittings and flanges vulnerable to corrosion must be protected by the manufacturer who will in particular take responsibility for draining, rinsing, drying, protecting and any adapted protection needed for transportation and storage in safe conditions.

1.2 Mechanical protection

The immobilization will be made by inserts appropriate to the dimension of the pipes, fittings and flanges.

**ATTACHMENT 4 POSITION OF SHIPPING-CARE MARK
CENTER OF GRAVITY MARK**



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ATTACHMENT 5 EXAMPLE OF TRANSPORT DRAWING

